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ORIGINAL RESEARCH

Policy information uncertainty and foreign institutional investors trading behavior: evidence from India

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Abstract
Using transaction–level data and exploiting a credible quasi–natural experiment set provided by India's unexpected tax–related policy announcement, we examine the impact of policy–related information uncertainty (PIU) on the trading behavior of foreign institutional investors (FII). Based on transaction–level data, we compute the daily net equity trading by all FII of each listed Indian firm. We find that FII reacts quickly and withdraws significant investments during the PIU period. However, the investments of the FII do not become positive immediately when policymakers eliminate the relevant PIU. Further analysis reveals that the impact of PIU subdues and FII equity flow returns to the pre–PIU level in the long term. The results suggest that policymakers, particularly in capital–constrained emerging markets, should be highly conscious of generating PIU if they wish to attract and retain overseas investors.

Keywords Policy–related information uncertainty · Foreign institutional investors · Emerging markets · Transaction–level data · Quasi–natural experiment

JEL classification G11 · G18

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1 Introduction

The information asymmetry (IA) hypothesis, which suggests that foreign institutional investors (FII) are less informed than domestic investors about host–country investments, is currently the dominant economic explanation for the trading behavior of FII in emerging markets. Since FII are less informed, IA studies argue that these investors primarily seek information from recent asset prices. The IA view thus concludes that the trading of foreign portfolio investors is driven by observed changes in recent asset prices (Brennan and Cao 1997; Griffin et al. 2004).

Several empirical studies have investigated the trading behavior of FII in emerging markets, primarily using non–structural dynamic estimations. In this context, by examining the joint dynamics of capital flows and equity returns, particularly in emerging Asian markets, several papers provide strong empirical support for the IA hypothesis in various forms where foreign investors are engaged in positive feedback trading (Bohn and Tesar 1996; Choe et al. 1999; Stulz 1999; Froot et al. 2001; Kim and Wei 2002; Griffin et al. 2004; Richards 2005). Recently, Ülkü (2015), using emerging European data, also found evidence of negative feedback trading. Another strand of literature often cites U.S. interest rates as a proxy for world interest rates and world risk aversion as an important factor influencing portfolio flows into emerging markets (Chuhan et al. 1998; Ülkü 2015; Sarno et al. 2016; Pagliari and Hannan 2024). However, in contrast, Bekaert et al. (2002) do not report any significant effect on capital flows to emerging markets due to an unexpected reduction in world interest rates.

In this study, we extend the literature by arguing that although the IA–related hypothesis is informative, it does not entirely explain the trading behavior of FII in emerging markets. We investigate whether policy–related information uncertainty (henceforth PIU) also describes the trading behavior of FII in emerging markets. We define PIU as the ambiguity associated with the scope and practicalities of implementing unexpected policy announcements by the host government.

We extend the literature on the trading behavior of FII in emerging markets by investigating how an unexpected and unfavorable policy announcement that creates an ambiguous investment environment explains the trading activities of FII. The unexpected and unfavorable policy announcement is related to a unique political risk for foreign investors. For example, Stulz (2005) argues that for foreign investors, host country characteristics are important due to the risks of expropriation by corporate insiders and the host government itself. He documents that the host state can establish, enforce, and break rules that affect investors' payoff within that country, including that of foreign investors. Similarly, Kerner (2014) suggests that if policy measures are not conducive (and clear) to foreign investors, they can react by boycotting the market and investing elsewhere where policy convergence is more advantageous. This again reflects the notion that the actions of policymakers themselves could disadvantage their competitive appeal in attracting foreign investors, who have the flexibility and choice to divert their funds across the globe.

Studies suggest that, although prices in financial markets are projections of future cash flows, they are also influenced by government actions (Bond and Goldstein 2015). This implies that the information foreign investors extract from recent prices concerning future cash flows is endogenous and could also be influenced by government policies, particularly by exogenous policy announcements. Bond and Goldstein (2015) further stress that these effects could be severe if the information and policy goals are not transparent. Although there is anecdotal evi-

dence of the impact of information uncertainty on the sentiments of foreign portfolio investors in emerging markets, the current literature lacks a systematic and rigorous scientific investigation. The lack of evidence may be attributed to the unavailability of reliable market–based data and an experimental setting for establishing a credible causal link between information uncertainty and the trading of foreign portfolio investors. We utilize unique transaction–level data and a quasi–experimental setting, where exogenous information uncertainty related to tax provisions is leveraged to establish a causal link with FII transactions.

Our setting of a sudden unexpected exogenous policy announcement is the General Anti Avoidance Reform (GAAR) declared in March 2012 by the Indian government, which threatened to impose additional tax obligations on FII transactions.[1] The announcement was unexpected and needed to be more transparent regarding its implementation. It needed clarity on how (whether the onus would be on FII or the tax authority) and to what extent (retrospectively or prospectively) the GAAR provisions would be applied. This created uncertainty in the investment environment for FII. Although a further announcement was made, postponing the effective date by one additional year, more clarity was needed regarding the ambiguities associated with the first announcement. Finally, at the end of June 2012, a draft guideline was issued that significantly clarified the scope and practicalities of the GAAR provisions.[2] The exogenous shock between March and June 2012, combined with the availability of unique daily transaction–level data on foreign equity portfolio trading, provides us with an ideal experimental setting to test how FII react to an ambiguous regulatory environment from PIU.

This paper has three important findings. First, FII reacts severely to the announcement of the GAAR, with positive FII flows before the announcement turning negative in the post–announcement period. This result supports our view that PIU negatively impacts FII's trading behavior. The PIU generates potential deadweight costs for the FII, who react by withdrawing from the market. Concerning economic significance, our results show that the average daily withdrawal by FII during the PIU period was at 0.543 basis points of the market capitalization (around INR 6.91 million or USD 0.128 million) for an average traded equity.[3] Second, the policy announcement's negative impact is immediate, observed from the day following the date of the first announcement, rather than from the proposed effective date. Third, the sentiments of the FII gradually turn positive when the PIU (ambiguities surrounding GAAR) are removed.

In summary, our study finds strong support for the conjecture that PIU, in addition to IA, influences the trading behavior of FII in emerging markets. These results are robust to a quasi–natural experiment and a series of robustness tests. Additionally, although this is not the focus of our study, we also find significant evidence of return–chasing behavior, where recent higher returns drive higher foreign inflows (consistent with the IA framework of Brennan and Cao 1997; and prior empirical evidence). These results complement the existing literature by providing credible evidence based on firm–level IA proxies, in contrast to existing studies that generally use market–level aggregate flow data (Bekaert et al. 2002; Ülkü 2015).

Our study makes two important contributions to the literature. First, we suggest PIU is a contributory driver in FII's trading behavior. We, therefore, add to the growing literature that

[1] See Sect. 2 for further elaboration on GAAR provision.

[2] See *The Hindu* (30 June 2012): "GAAR norms only a draft, says PMO" URL: https://www.thehindu.com/business/Economy/gaar–norms–only–a–draft–says–pmo/article3584189.ece.

[3] Applying the daily average USD/INR rate for the year 2012 of INR 54/US $ (Source: India's Central Bank, Reserve Bank of India).

investigates the trading behavior of FII in emerging markets where IA is the core explanatory factor (Brennan and Cao 1997; Richards 2005; Marshall et al. 2022; Abou Tanos and Jimenez–Garcès 2022). Bowe and Damuta (2001) also provide convincing evidence that the impact of FII information on asset prices in Asian markets is enhanced during crisis periods. In a recent study, Hao and Li (2024) demonstrate that the IA related to economic policy uncertainty has a negative impact on both the domestic and cross–border flows of portfolio investments. Likewise, Francis et al. (2021) find a reduction in institutional investors' holdings during the period surrounding political elections.

Second, we contribute to the literature by using individual firm and FII transaction level data to capture the exogenous effect of PIU (Marshall et al. 2022). Prior studies on PIU have predominantly relied on aggregate market–level flow data, typically measured at monthly, quarterly, or annual intervals (Griffin et al. 2004). However, such aggregate–level analyses may obscure underlying micro–level heterogeneity in investor behavior. They may also suffer from limitations in statistical power, which may contribute to the mixed empirical findings documented in the literature on investor trading dynamics (Froot et al. 2001). Regarding cross–border capital flows, several studies on foreign direct investment (FDI) indicate that FDI inflows decline during periods of uncertainty and recover once the uncertainty is resolved (Julio and Yook 2016; Choi et al. 2020). These studies may suffer from endogeneity. For instance, aggregated FDI flows may fail to capture the impact of unobserved factors such as short–term market conditions and policy risks. Our study differs in that we examine dynamic daily portfolio flows of FII. The granularity of the dataset affords us the statistical power to undertake very short–term regression analysis (Marshall et al. 2022). More importantly, instead of using prices in a simple OLS framework, which can be endogenous, we employ a quasi–natural experiment by exploiting an unexpected exogenous shock, thereby offering the opportunity for a quasi–natural investigation. Using such an exogenous event strengthens causal inference (Angrist and Pischke, 2009), as it provides an opportunity to disentangle the impact of PIU from confounding market dynamics.

The rest of the paper is as follows. The next section provides a brief review of the background and key dates related to the GAAR announcement. Section 3 describes the theoretical framework we use to motivate our empirical study. Section 4 describes the FII transaction data and discusses the empirical results in Sect. 5. Finally, we conclude the paper in Sect. 6.

2 General anti–avoidance reform (GAAR)

On March 16, 2012, India's Finance Minister unexpectedly announced the introduction of GAAR provisions on FII effective from April 1, 2012. GAAR was an anti–tax evasion provision aimed at empowering Indian tax authorities to deny capital gains tax benefits if the transaction lacked commercial substance or was executed solely for tax benefits (see Appendix A for information on the potential tax benefits enjoyed by FII in India).[4] However, as shown in Fig. 1 (box 1), two significant uncertainties were associated with the announce-

[4] "The government wants to plug as many holes as possible," said Jagannadham Thunuguntla, head of SMC Global. "If your fund actually belongs to the US and just to get a tax advantage, you've created an artificial company in Mauritius…Mauritius is not being used as a residence, just a tax shelter. [The government's point is that] the tax–avoidance agreement with Mauritius should not be made a mockery of." (Financial Times, 29 March 2012).

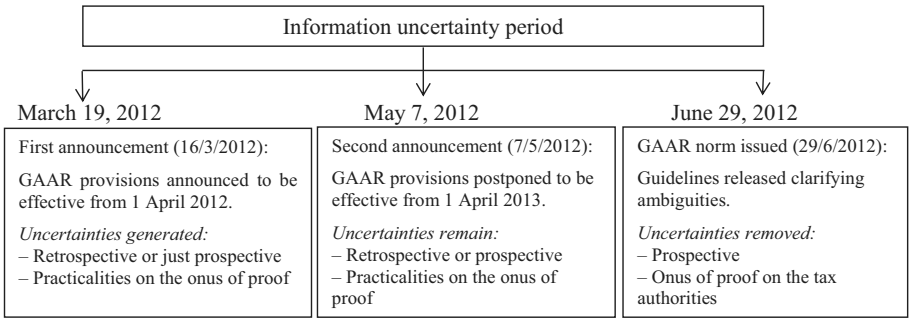


Fig. 1 GAAR uncertainty period timeline. This figure shows the different announcements related to GAAR and the associated uncertainties

ment. First, it was unclear whether the GAAR provisions would be imposed retrospectively or prospectively. Second, whether the FII themselves needed to prove that the transaction was not executed for tax benefit only, or the burden of proof would be on the tax officials.

On May 7, 2012, India's Finance Minister made a second announcement, notifying FII that GAAR would be delayed by one year, i.e., the effective date would be April 1, 2013, rather than April 1, 2012, as initially proposed in the first announcement. However, as noted above, the second announcement did not clarify the two uncertainties related to the first announcement. The first uncertainty was whether GAAR would be invoked prospectively or retrospectively. The second was whether the onus of proof would be that of the tax authorities or FII (see box 2 of Fig. 1). Finally, on June 29, 2012, the government clarified the ambiguous provisions, indicating that GAAR would be imposed prospectively on transactions from April 1, 2013, i.e., the proposed implementation date, and the onus of proof would be on the tax authorities (see Box 3, Fig. 1).[5] As seen in Fig. 1, the period between the first trading day (March 19 2012) following the first announcement (March 16 2012) and the third announcement (June 29 2012) offers us an ideal experimental set–up to investigate how FII reacts to the PIU created by the sudden and unexpected announcement of the GAAR provision.[6]

3 Hypothesis development

Our study follows the theoretical framework of Bacchetta and Van Wincoop (2000). This framework demonstrates the dynamics of capital flows in emerging markets when gradual liberalizations (such as tax reductions) are introduced. Suppose FII prefers to allocate their wealth, *W*, between the Indian market *(IND)* and other *N* countries. Thus, the total number of markets invested is $N+1$. Period *t* expected returns on investment in each (n_i) of the other countries ($N = \sum_{i=1}^{N} n_i$) are given by $r_{n_i t} \sim N\left(\overline{\mu}_N, \sigma_N^2\right)$. Let period *t* returns on Indian equities *(IND)* be $\mu_{IND,t} \sim N(\mu_{IND}, \sigma^2_{IND})$. The return on the Indian market is subject

[5] See *The Hindu* (30 June 2012): "GAAR norms only a draft, says PMO".

URL: https://www.thehindu.com/business/Economy/gaar–norms–only–a–draft–says–pmo/article3584189.ece.

[6] From information available on 7 January 2015, GAAR has been postponed to 2017. See http://www.livemint.com/Politics/6IP8PO6wHsvQe7FkjFaruK/GAAR–deferred–by–two–years.html (date accessed 7 January 2015).

to a tax, $\tau_{IND,t}$ for foreign investors, including FII. This tax captures the various barriers or costs to investments faced by FII, such as capital controls and taxation. An increase in $\tau_{IND,t}$ is considered as deliberalization. The incorporation of the tax means a net pay–off of:

$$r_{IND,t} = \mu_{IND,t} - \tau_{IND,t} \tag{1}$$

where, $\tau_{IND,t}$ represents the existing applicable taxes, which the investors know in advance. The conditional expected return after tax is $\bar{r}_{IND,t} = \bar{\mu}_{IND,t} - \tau_{IND,t}$, where $\bar{\mu}_{IND,t}$ is the conditional expected return. The information set in our case includes information asymmetry and other push and pull factors. For simplicity, we assume that the returns are uncorrelated across countries and $\sigma^2_{IND} = \sigma^2_N$. Assuming that investors have a preference for exponential utility function $U(C) = e^{-\theta C}$, and given that consumption C is equivalent to portfolio return, i.e. $R_t \times W$, and θ is the degree of risk preference ($\theta > 0$), FII prefers allocations that maximize period t utility, which is a function of the mean–variance trade–off as presented in Eq. 2:

$$max_{\alpha_{nt}} E(R_t) - \frac{\gamma}{2} var(R_t) \tag{2}$$

where $n \in [1, N+1]$, India is the $(N+1)^{st}$ equity market, α_{nt} is the allocation weight of country n_i in the portfolios (with $\sum_{n=1}^{N+1} \alpha_{n_i t} = 1$), $\gamma = \theta W$,. The portfolio return R_t is given by Eq. 3:

$$R_t = \sum_{n=1}^{N} \alpha_{n_i t} . r_{n_i t} + \alpha_{N+1,t} . r_{IND,t} \tag{3}$$

Defining the average expected return on other countries as $\bar{r}_t = \sum_{n=1}^{N} \bar{r}_{n_i t}/N$ and the expected return on the Indian market as $\bar{r}_{IND,t}$; the solution of the objective function leads to an allocation in the Indian market (i.e. $(N+1)^{st}$ market) by FII of:

$$\alpha_{N+1,t} = \frac{1}{N+1} + \frac{\bar{r}_{IND,t} - \left[\frac{\bar{r}_{IND,t} + N\bar{r}_t}{N+1}\right]}{\gamma \sigma^2} \tag{4}$$

Using $\bar{r}_{IND,t} = \bar{\mu}_{IND,t} - \tau_{IND,t}$, we get:

$$\alpha_{N+1,t} = \frac{1}{N+1} + \frac{(\bar{\mu}_{IND,t} - \tau_{IND,t}) - \left[\frac{\bar{r}_{IND,t} + N\bar{r}_t}{N+1}\right]}{\gamma \sigma^2} \tag{5}$$

Following the arguments of Bacchetta and Van Wincoop (2000), and Edison and Warnock (2008), Eq. 5 suggests that an expected increase in $\tau_{IND,t}$ should affect the future payoff and thus discourage FII from reducing their portfolio weight in the Indian equity market. In our case, the unexpected and ambiguous announcement of the GAAR provision should increase the expected $\tau_{IND,t}$ to FII. If the probability–weighted expected value of the additional tax liability is $\phi_{IND,t}$ and with the assumption of zero pre–announcement liabil-

ity, the expected liability $\phi_{IND,t}$ increases from $\phi_{IND,t} = 0$ to $\phi_{IND,t} = \bar{\phi}_{IND,t}$. After the announcement, the portfolio weight in the Indian equity market (i.e. (N+1) st market) by FII is thus given by $\widehat{\alpha}_{N+1,t}$ as:

$$\widehat{\alpha}_{N+1,t} = \frac{1}{N+1} + \frac{(\bar{r}_{IND,t} - \bar{\phi}_{IND,t}) - \left[\frac{\left(\bar{r}_{IND,t} - \bar{\phi}_{IND,t}\right) + N\bar{r}_t}{N+1}\right]}{\gamma\sigma^2} \tag{6}$$

Equation 6 implies that an expected rise increases in $\phi_{IND,t}$ from 0 to $\bar{\phi}_{IND,t}$, after the GAAR announcement, should result in outflows from the Indian equity market. Subtracting Eq. 6 from Eq. 5 shows us the difference in portfolio allocation in the Indian market after the announcement of GAAR provisions:

$$\alpha_{N+1,t} - \widehat{\alpha}_{N+1,t} = \bar{\phi}_{IND,t}\left(\frac{N}{\gamma\sigma^2}\right) \tag{7}$$

Thus, the difference in allocations, as shown in Eq. 7, is attributable to the increase in tax liability $\bar{\phi}_{IND,t}$ arising from the policy announcement that created uncertainty. The literature identifies such friction as a country's political risk. Bekaert and Hordick (2018) note that for a foreign investor, such a risk could be the possibility that the host state could dramatically change the "rules of the game", which, among many other issues, also includes an unexpected increase in taxes.

Based on the model, we would expect portfolio inflows to follow its pre–announcement trend (keeping $\tau_{IND,t}$ constant) but expect outflow with the introduction of new tax liability $\bar{\phi}_{IND,t}$ for FII. Therefore, we propose testing the following hypothesis.

H_1: In the post–GAAR announcement period, we expect firms to experience significant withdrawal from FII compared to pre–GAAR period.

Further, we would expect the $\bar{\phi}_{IND,t}$ to keep increasing as the uncertainty related to the announcement increases, and thus, in a dynamic case, we would expect the outflow to continue until $\bar{\phi}_{IND,t}$ is eliminated, which, in our case, is the third announcement that eradicates the ambiguities related to the GAAR announcement. This argument is consistent with existing literature, which finds reduced equity capital flow during high–uncertainty periods (Agoraki et al. 2024; Kwabi et al. 2023; Marshall et al. 2022; Frobes and Warnock, 2012; Rey 2018). Hao and Li (2024) and Wang et al. (2024) demonstrate that economic policy uncertainty deters FII's investments due to high information asymmetry. Choi and Havel (2025) show the equity and the debt capital flow declines when the geopolitical risk is very high in emerging markets, which is mostly driven by poor institutional quality and closed capital market. Glossner et al. (2025) and Neupane et al. (2024) investigate the COVID–19 crisis period and find a significant withdrawal of capital by institutional investors, particularly by foreign investors, from the equity market due to liquidity concerns raised by the crisis.

However, findings in the literature suggest that FII investments may not flow in promptly when uncertainty is removed, particularly in the case of emerging markets, due to what is referred to as the lingering effect of uncertainty. The lingering effect of uncertainty refers to the prolonged influence of uncertainty on economic decision–making, persisting even after the initial source of uncertainty has been resolved. This effect arises as FII may adopt a cautious approach, delaying investment, due to the costs associated with adjusting expectations about policy changes. Empirical evidence suggests that uncertainty shocks induce a "wait–and–see" behavior, leading to a sustained drag on economic activity and financial market volatility (Bloom 2009). This lingering effect may be exacerbated due to incomplete information about policy announcements. Bachetta and van Wincoop (2000) argue that there may be uncertainty surrounding the extent and success of policy reforms for foreign investors in emerging economies. Although the government may announce that it has adopted reforms, such as eliminating $\bar{\phi}_{IND,t}$, the questionable government's credibility issues in abruptly changing policies in emerging markets may not fully convince the FII (Bachetta and van Wincoop, 2000). As such, until FII fully updates its perception of the third announcement by the government that eradicates the ambiguities surrounding the GAAR, capital inflows may be lower compared to periods before the PIU.

Empirical evidence provides support for the argument. Gulen and Ion (2016) document that investment flows may take, on average, two to three years to recover after policy uncertainty. Likewise, Julio and Yook (2012) and Jens (2017) also find evidence that supports the conjecture that investments do not completely recover in the post–uncertainty periods (elections). Honig (2020) offers similar evidence, which shows that foreign capital outflows before elections but do not revert to inflows immediately after the election. Marshall et al. (2022) also find that the elimination of threats by the government does not result in a sudden and significant inflow of capital, unlike the exit reaction to tax threats.

Thus, in light of the existing literature on the lingering effects of uncertainty, we propose testing the following hypothesis.

H_2: In the post–GAAR eradication period, FIIs' capital inflows should either remain negative or, if positive, should be significantly smaller in magnitude compared to the outflows after the GAAR announcement period.

4 Data and preliminary figures

The primary dataset used in this study is the transaction–level data of FII and their subaccounts. The Securities and Exchange Board of India (SEBI) maintains a record of individual FII transactions available from National Securities Depository Limited (NSDL).[7] We restrict our analysis to the purchase and sale of listed equities in the secondary markets.

[7] NSDL reports the following nineteen different fields of this dataset: Custodian code, Report Date, Transaction ID, FII Registration (identify masked), Sub Account Registration Number (identity masked), Broker Registration Number (identify masked), Scrip name, ISIN code, Transaction date, Transaction type, Stock exchange code, Settlement code, Transaction rate, Transaction quantity, Value, Instrument type, Reason for delay in reporting, Reporting type, and Reason for amendment. For further details see the website of NSDL (https://www.fpi.nsdl.co.in/web/StaticReports/FIITradeWise2008/FIITradeWise2008.htm).

These comprise 99.84% of all transactions by FII. For instruments traded, 99.94% of them were listed as equities, which we included in our analysis. We cannot identify the details of individual FII, their sub–account, and the corresponding broker of the related transaction (see footnote 10). Hence, we investigate the aggregate net trading by all FII in individual equities using firm–level equity identifications based on the International Securities Identification Number (ISIN) code. Given our purpose of testing whether the PIU period (March 19 – June 29, 2012) affected the trading behavior of FII, our sample covers the period from January 1 to June 29, 2012.

Figure 2 reports monthly net trading (purchase – sales, in INR million) of the listed Indian equities by all FII for 2012. The total net value traded by FII for the first three months registered positive inflows, with the highest being INR 217 billion for February 2012. However, after the GAAR announcement on March 16 2012, these aggregated net values for April and May 2012 were very low (negative in May 2012). Although the net trading increased in June 2012, after the postponement announcement of May 7 2012, the value of roughly INR 40 billion for June 2012 is almost half that of the figure observed for March 2012. The figures for April, May, and June 2012 are the lowest of any other months in 2012. We attribute this decrease to the PIU period from March 19 – June 29, 2012.

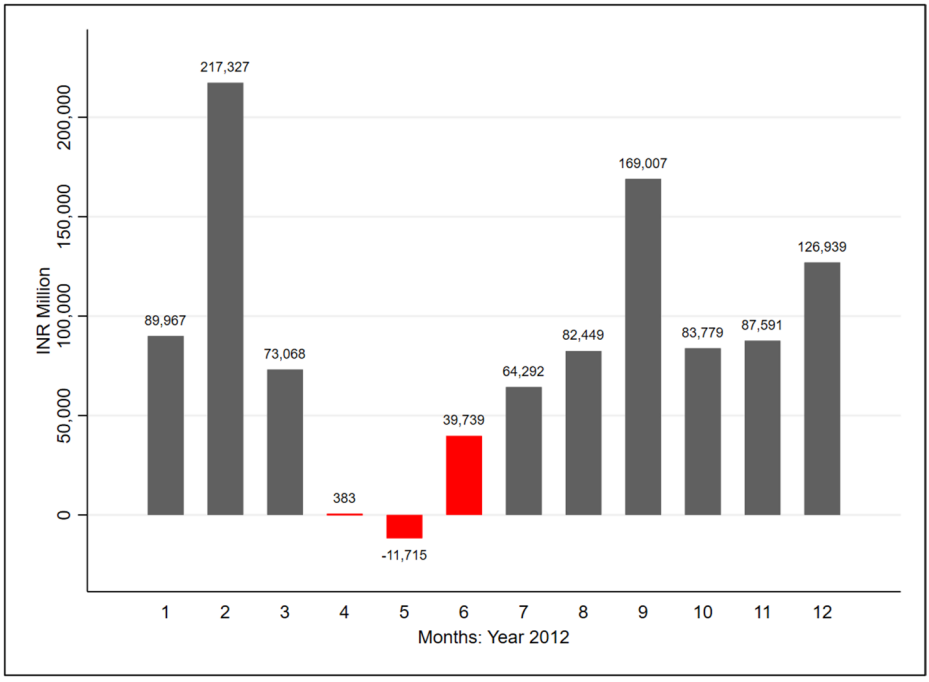


Fig. 2 Month–wise total net trading (inflows/outflows) by foreign portfolio investors. This figure shows the monthly total net trading (purchase – sales) in Indian equities by foreign institutional investors for 2012. Bars with positive figures reflect net inflow and negative net outflow. The red bars denote the period of heightened GAAR uncertainty. Data Source: Securities Exchange Board of India

5 Empirical results

5.1 Summary figures

Following the literature on modeling the dynamics of trading by foreign investors in emerging markets (Bekaert et al. 2002; Froot et al. 2001; Richards 2005; Marshall et al. 2022), we aggregate daily FII trading at the firm level and scale the net trading by the firm's previous day's market capitalization. We denote this variable as $NTSMC_{it}$, which is net trading scaled by the previous day's market capitalization (in basis points) as defined in Eq. 8:

$$NTSMC_{it} = \frac{\sum_j (quantity_{j,i,t} \times price_{i,j,t})}{MCap_{i,t-1}} \tag{8}$$

where: $\sum_j (quantity_{j,i,t} \times price_{i,j,t})$ is the net purchase by all FII on trading day *t* of equity *i*. The term $quantity_{j,i,t}$ is the number of equity shares *i* traded by FII *j* on trading day *t*. The value of $quantity_{j,i,t}$ is positive if the transaction is a purchase and negative if it is a sale. The term $price_{i,j,t}$ is the price paid by FII *j* for equity *i* on trading day *t*. The term $MCap_{i,t-1}$ is the market capitalization of firm *i* on trading day *t–1*.

We examine the summary figures of five different window periods and conduct a simple paired *t*–test for the difference in $NTSMC_{it}$ before and after the GAAR announcement on March 16, 2012. For example, for the one–day window, the mean of $NTSMC_{it}$ for Pre–GAAR uses data for March 16 2012, and the Post–GAAR includes data for March 19 2012. Similarly, for the one–week window, we find the mean of $NTSMC_{it}$ from 12 to 16 March (Pre–GAAR week) and 19–23 March (Post–GAAR week). This suggests that for the one–week window period, the sample contains two weeks of data. The same applies to the one– and two–month windows. For the GAAR uncertainty window, the post period spans the uncertainty period, i.e., from March 19 – June 29 2012. The pre– PIU period includes a sample from January 1 – March 16 2012.

Table 1 shows that the average daily $NTSMC_{it}$ is negative and statistically significant at the 1% level in the Post–GAAR period, relative to the Pre–GAAR period. We observe a statistically significant outflow of 0.549 basis points in one trading day following the GAAR announcement. The average $NTSMC_{it}$ difference figures are very similar in magnitude for the one–week, one–month, and two–month windows and the GAAR uncertainty period.[8] We continue to observe a statistically significant outflow of −0.543 basis points during the GAAR uncertainty period. With the average market capitalization being INR 127,157.4 million, the economic effect translates into a daily average of INR 6.91 million (USD 0.128 million) per equity during the GAAR uncertainty period.[9] These difference figures signal that the uncertainty created by the abstruse GAAR provisions may have had detrimental consequences on the trading sentiments of FII.

[8] Instead of using firm level data the results of paired *t*–test are similar when we use daily aggregated time series data.

[9] The average USD/INR rate was 54/USD for the year 2012 (Source: India's Central Bank, Reserve Bank of India).

Table 1 Average daily net equity trading

Window periods	Pre–GAAR I	Post–GAAR II	Difference III = II – I	t–stat	*p*–value	Observations
1–Day	0.539	–0.010	–0.549***	–2.791	0.001	773
1–Week	0.813	0.248	–0.565***	–4.086	0.000	3,877
1–Month	0.354	–0.044	–0.398***	–5.323	0.000	13,843
2–Months	0.354	–0.136	–0.489***	–7.994	0.000	22,958
GAAR Uncertainty Period	0.343	–0.200	–0.543***	–12.362	0.000	42,470

The table shows the paired *t*–test of the differences in average daily net trading value scaled by the previous day's market capitalization (in basis points) of listed stocks on the Indian stock market by all FII. Except for the *GAAR Uncertainty Period*, all other window periods denote the different periods of paired *t*–tests. *Pre–GAAR* shows the average value for the corresponding window before the GAAR announcement date (March 16 2012) and the *Post–GAAR* period after the announcement date. For example, for the one–day window, the *Pre–GAAR* period is March 16 and the *Post–GAAR* period is March 19 2012. The pre–window of the *GAAR Uncertainty Period* ranges from January 1 – March 16 2012, and the post–period ranges from March 19 – June 29 2012. The *difference* shows the Difference between *Post–GAAR* and *Pre–GAAR* average values. *t–stat* is the t–statistics of the *Difference* figure with the probability of the alternative hypothesis that the average difference is less than zero (i.e., Post–GAAR average – Pre–GAAR average < 0) denoted by *p–value*. *Observations* is the sample size included in each window.

5.2 Basic regression analysis

The first set of daily panel data regressions we run is based on different periods, ranging from a one–day to a two–month window. We create a dummy ($GAAR\,effect_t$) for the second period of each window. For example, for the one–day window, the $GAAR\,effect_t$ dummy takes a value of one for the second trading day (i.e., March 19 2012) of the announcement period of March 16 2012 (March 17 and 18 are the weekend). The sample thus includes only two trading days of data, i.e., 16 and 19 March 2012. We generate similar dummies for other window periods.

Following the existing literature, we control for several factors ($\boldsymbol{X}$ – defined in Appendix B Table 10) to rule out any other rival explanations that the shock–based exogenous $GAAR\,effect_t$ variable may be competing (Marshall et al. 2022). We lag all controls used by one day (where unavailable, we use the most recent information). The first control we incorporate is at the firm level.[10] As noted earlier, tests of IA suggest that recent returns have a significant influence on foreign investors in emerging markets (Brennan and Cao 1997). We capture this by taking the previous day's average return of the firms in the FII trade on a particular day. This factor, sourced from the Prowess database, captures the momentum effect (the IA hypothesis) and any other firm characteristics or news that might motivate FII to trade. We denote this as *Firm return* when reporting the results.

The second set of controls we use is time–varying national factors that may influence investors' trading decisions. The literature notes these as pull factors (Eguren–Martin et al. 2024; Griffin et al. 2004; Hau and Rey 2006). For the market–level momentum hypothesis (Griffin et al. 2004), we use the one–day lagged return on India's largest and most liquid stock exchange – the National Stock Exchange of India (*NSE return*). We obtain the NSE 500 index from the NSE website. High local equity market volatility may negatively impact FII equity flows (Ülkü 2015). We calculate equity market volatility as the daily standard deviation estimated using the past 90 days' return on the NSE 500 index (*NSE volatility*). To account for the exchange rate movement effect, we include the USD/INR daily exchange

[10] Firm size is already considered in the construction of the dependent variable noted in Eq. 1.

Table 2 Summary statistics of controls

	Mean	Median	Std. Dev	Min	Max
Firm return (%)	0.266	0.180	2.013	–3.294	4.316
NSE return (%)	0.103	0.150	0.947	–1.562	1.744
NSE volatility (%)	1.191	1.214	0.097	0.907	1.301
USD volatility (%)	0.625	0.614	0.066	0.532	0.726
Real GDP growth rate (%)	3.579	3.193	0.391	3.193	3.974
EM return (%)	0.015	0.027	0.870	–1.430	1.433
World return (%)	0.041	0.115	0.732	–1.329	1.384
US TB return (%)	0.172	0.180	0.028	0.120	0.210
Global VIX return (%)	–0.101	0.000	5.495	–10.326	10.414

This table shows the summary statistics of the control variables used in this study during our sample period from January 1 2012 – June 29 2012. All the variables are defined in Appendix B Table 10.

rate volatility (standard deviation) constructed using the exchange rate figures from the previous 90 days obtained from the Reserve Bank of India, denoted as *USD volatility* (Hau and Rey 2006). The time–varying national macroeconomic prospect is factored in by the recent availability of last quarter's *real gross domestic product (GDP) growth rate* sourced from the OECD.

Finally, we also control global information variables, referred to as push factors (Eguren–Martin et al. 2024; Richards 2005). We use the previous day's return on the world market (*World return*) using the MSCI All–Country Total Return Index. We also use the previous day's return on the emerging market (*EM return*) using the daily MSCI Emerging Markets' total return index. The MSCI return indexes are obtained from Thomson Reuters. Global risk aversion is incorporated by using the lagged return of the VIX index of the Chicago Board Options Exchange (*Global VIX return*) as a measure of the market's expectation of near–term global volatility (Porras and Ülkü 2015).

Furthermore, we also incorporate the *flight–to–safety* effect by using the previous day's return on one–year U.S. Treasury bills (*U.S. TB return*) rates acquired from the U.S. Treasury International (Sarno et al. 2016). This captures any changes in the global interest rates that could persuade FII to transact in risky emerging markets.[11] Table 2 presents the summary statistics of the control variables during our sample period.

The daily data available for each firm allows us the statistical luxury of running a daily fixed effect panel data model, as shown in Eq. 9, controlling for firm fixed effects ($\boldsymbol{\gamma}_i$) as well as time (days) fixed effects ($\boldsymbol{\delta}_t$). Further, standard errors clustered at the firm level and time (days) level (where denoted):

$$NTSMC_{it} = \beta\,(GAAR_Effect_t) + \boldsymbol{X}.\delta + \boldsymbol{\gamma}_i + \boldsymbol{\delta}_t + e_{it} \tag{9}$$

Based on our theoretical framework, if the drop in FII trading on day one post the GAAR announcement is attributed to PIU, we would expect β to carry a negative sign and be statistically significant. The results of the various versions of Eq. 9 are presented in Table 3. Models I–IV show that the GAAR announcement significantly changes the trading behavior of FII, with a quantitative effect of, on average, a minimum of 0.419 basis points ($p<0.01$)

[11] In all our regression models, the Variance Inflation Factor (VIF) of the control variables is less than 10, suggesting that our empirical models do not suffer from multicollinearity issues.

Table 3 Different periods–based regressions

	One–day window	One–week window	One–month window	Two–month window	GAAR Uncertainty Period	GAAR Uncertainty Period
	I	II	III	IV	V	VI
$GAAR\,effect_t$	–0.517***	–0.581***	–0.419***	–0.595***	–0.533***	–0.473***
	(–2.78)	(–3.69)	(–3.79)	(–6.37)	(–6.39)	(–3.71)
Firm return						0.235***
						(13.41)
NSE return						–0.073
						(–1.19)
NSE volatility						–0.468
						(–0.48)
USD volatility						0.393
						(0.13)
Real GDP growth rate						–0.356
						(–0.65)
EM return						–0.104
						(–1.03)
World return						0.176
						(1.37)
U.S. TB return						7.350**
						(2.20)
Global VIX return						0.009
						(0.74)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effect (days)	No	No	No	No	No	Yes
Number of firms	319	505	645	749	812	812
Observations	638	3,798	13,771	29,047	47,461	47,461
Adjusted R^2 (within)	0.010	0.006	0.003	0.005	0.006	0.018
Adjusted R^2	0.252	0.225	0.157	0.144	0.139	0.154

This table reports different regression results of the following regression specification:

$$NTSMC_{it} = \beta\ (GAAR\,effect_t) + \boldsymbol{X}.\delta + \boldsymbol{\gamma}_{\boldsymbol{i}} + \boldsymbol{\delta}_{t} + e_{it}$$

where $NTSMC_{it}$ is the day t net trading value by all FII scaled by the previous day's market capitalization (in basis points) of listed stocks (i) on the Indian stock market. Firms traded are indexed as i, and daily periods are indexed as t. $GAAR\,effect_t$ is the dummy variable which takes the value of one in the post–GAAR period for one–day, one–week, one–month and two–month window periods. For example, for the one–day window, the *GAAR effect* dummy takes the value of one if the period is day 2 (i.e., March 19 2012) post–announcement period of March 16 2012 (17 and 18 are the weekend). The sample thus includes only two subsequent trading days of data, i.e., for 16 and 19 March 2012 for all FII. Similar dummies are generated for other window periods discussed in Table 1. $\boldsymbol{X}$ reflects the set of control variables. All the variables are defined in Appendix B Table 10. γ_i and δ_t is the vector of firm and time (day) dummies and controlling for firm and time fixed effects, respectively (where applicable). e_{it} is the error term. Standard errors are corrected for clustering at the firm (double clustered at the firm and time for model VI). *, ** and *** denote statistical significance at the 10%, 5% and 1% significance level respectively.

of market capitalization per day for an average equity during the uncertainty period. From an endogeneity point of view, it is worth noting that the estimate ($\beta = -0.519$, $p<0.01$) related to the shortest possible period, i.e., one–day window, affords us high confidence in avoiding the influence of other factors. This is particularly important in our setting, where the daily average per equity trading figures were positive (see Fig. 2) for the last three

months. However, the announcement on March 16 unexpectedly plunged the corresponding trading on March 19 into negative territory. The one–day window result also suggests that the effect of PIU takes place right from the day after the announcement (i.e., March 19 2012) instead of the effective day of April 1 2012.

Although the short windows indicate the GAAR effect, we are particularly interested in examining the impact of the *Post–GAAR Period* window, i.e., the period between March 19 2012, and June 29 2012. In the sample covering the first six months of 2012, we produce a dummy taking the value of one for the *Post–GAAR Period* in Models V and VI. This is the PIU period (see Fig. 1), during which FII faced increased deadweight costs of investing in the Indian market. However, given the length of the *Post–GAAR Period* window, we now need to be confident that our results are not driven by other alternative explanations that could be correlated with our *GAAR effect* dummy. In Model V, we include the *GAAR effect* dummy without controlling for other variables; in Model VI, we control for these variables. Since we run a fixed effect panel data model, we also control for firm–specific effects (γ_j) and time (days) fixed (δ_t) and correct the standard error by double clustering at the FII and time (day) levels (Peterson 2009).

Models V and VI demonstrate that our key variable of interest (GAAR effect) enters the regressions with the expected signs and is statistically significant at the 1% significance level. After controlling for all the determinants, we find a negative equity flow of −0.473 basis points ($p<0.01$) following the GAAR announcement. Economically, this translates to a daily withdrawal of INR 6.02 million (approximately USD 0.12 million) for each equity. The results support our central hypothesis that the sudden pull–out by FII is attributed to the ambiguous GAAR provisions announced by the Indian government, which created an environment of uncertainty. Among the controls, the consistent statistical significance of *Firm return* supports the IA arguments. However, we also find evidence of the significant influence of global push factors, particularly U.S. TB rates, which is consistent with recent studies (see Ülkü 2015; Sarno et al. 2016).

5.3 Firm–level difference in differences analysis

We have used various period dummies to reflect the GAAR effect in all our fixed–effects regressions. In this framework, we control for time–varying firm–level, emerging, and global market–level factors, as well as unobserved firm–level and time–fixed effects. Despite this, our estimate could still be driven by other systematic shocks that could have occurred during the same period, particularly in the absence of comparable treatment and control groups. We address this issue using the firm–level difference–in–differences (DiD) quasi–experimental method. Vig (2013) notes that DiD compares the effect of a shock on groups that are affected or more affected (henceforth, treatment) with those that are unaffected or least affected (subsequently, control). As such, when we take the difference in the differences between the treatment and control groups' average $NTSMC_{it}$, before and after the GAAR announcement, we eliminate the bias that stems from changes other than the GAAR effect that could potentially drive FII trading decisions.

The challenge in our case is identifying the treatment and control groups. As noted earlier, one of the uncertainties associated with the GAAR announcement was that it was unclear if the new provisions would be applied to retrospective or prospective transactions. Therefore, it is conceivable that firms on which FII had relatively higher exposure of cumu-

lated net trading in the pre–announcement period would be more affected than those with lower exposure. The Prowess dataset reports information on the two–digit National Industrial Classification of India (NIC2). As shown in Appendix C Table 11, we use the NIC2 code and calculate the sum of the FII net equity trading figures of all firms in the different sectors for January – April 2012. We sum up the figures for January and February to identify the cumulated net trading exposure for the past two months before the GAAR announcement and sort the entire table against this total figure reported in column 4.

Column 4 of Appendix C shows that the cumulated net trading exposure in certain sectors, such as financial, IT, and manufacturing, is significantly higher than in coal mining and communications sectors. We use the median net trading figure of approximately INR 189 million to create the treatment and control groups. The sectors with a cumulated net trading exposure of INR 189 million or more are in the treatment group, while those with an exposure below this amount are in the control group.[12] We argue that any adverse shock in FII sentiment should have a more negative impact on the treatment group firms than on the control group firms.

To examine the parallel trend assumption, we examine the weekly cumulative net traded value (weekly purchase value – weekly sales value) for the treatment and control groups for the pre–GAAR (week 1–11) and post–GAAR periods (week 11–25). As seen in Fig. 3, both the treatment and control group exhibits a similar upward trend in the cumulative net equity trading value before the GAAR announcement week (denoted by the horizontal line). However, in the post–GAAR announcement period, while the control group firms showed no significant changes in cumulative net equity trading, the treatment group firms experienced a downward trend. The cumulated net trading value is INR 370 billion for the treatment group during the pre–GAAR period compared to INR −8.4 billion (sales greater than purchases) for the control group. However, in the post–GAAR period, the drop in the treatment group is significantly higher (54–370 = INR −316 billion) compared to a slight increase in the control group [–6.8 – (–8.4) = INR 1.6 billion].

We now perform the DiD examination in two ways. First, we evaluate the mean difference in $NTSMC_{it}$ for the treatment and control groups before and after the GAAR announcement. As reported in Panel A of Table 4, the average $NTSMC_{it}$ for the treatment group during the pre–GAAR period was 0.402 basis points, whereas the same for the post–GAAR period was −0.164 basis points, with the difference being −0.566. This difference, statistically significant at the 1% significance level, implies a substantially material downward shift in the trading trend by FII. The same difference for the control is 0.147 basis points. However, this difference is also not statistically significant. This further suggests a comparatively much smaller GAAR effect on the control group. The mean DiD is, therefore, −0.713 basis points per day of market capitalization for a typically listed firm on the Indian stock market during the PIU period. Economically, this translates into a daily withdrawal of INR 9 million for each equity (around USD 0.17 million).

The second approach we take to use the DiD method is by running regressions with different specifications of Eq. 10 below:

$$NTSMC_{it} = \beta\ (GAAR\,effect \times\ TRMT_GRP) + \boldsymbol{X}.\delta + \boldsymbol{\gamma}_{\boldsymbol{i}} + \boldsymbol{\delta}_{t} + e_{it} \quad (10)$$

[12] Given the wide variations in the dataset we use the median value. However, the use of mean value produces qualitatively and quantitatively similar results.

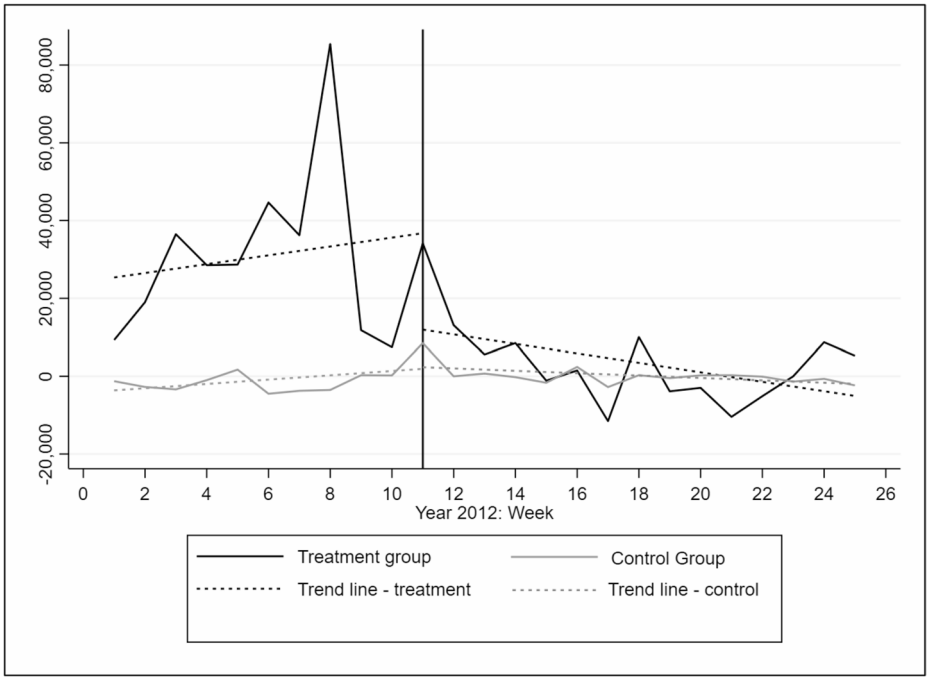


Fig. 3 Weekly cumulated net transaction value pre– and post–GAAR period. This figure shows the treatment and control group's weekly cumulated net trading value (Purchase – Sales) for foreign institutional investors for the pre–GAAR and post–GAAR periods. The pre–GAAR period runs from January 1 – March 16 2012 (Week 1 – Week 11), and post–GAAR runs from March 19 – June 29 2012 (Week 11–Week 25). The dotted line shows the trend line, and the horizontal line represents the week ending March 16 2012 (Week 11)

where $GAAR\,effect$ is the dummy variable that takes the value of one for *Post–GAAR Period* from March 19, 2012, to June 29, 2012, and zero for the *Pre–GAAR Period* from January 1, 2012, to March 16, 2012. $TRMT_GRP$ is the dummy which takes the value of one if the firm falls within the treatment group (as shown in Appendix C Table 11) based on two–digit NIC sector classification. The estimated effect (β) reflects the extent to which the *Post–GAAR Period* deviates from the treatment firms' $NTSMC_{it}$, relative to control firms and the *Pre–GAAR Period*, following the announcement made on March 16, 2012. Based on our theoretical framework, we expect β to be negative and statistically significant. $\boldsymbol{X}$ denotes the set of control variables, as mentioned earlier, $\boldsymbol{\gamma}_i$ is firm fixed effects and $\boldsymbol{\delta}_t$ controls for time–fixed effects (days). Standard errors are double clustered at the firm and time (days) levels.

Panel B of Table 4 shows that the $GAAR\,effect \times TRMT_GRP$ variable enters all regressions with an expected negative sign and is statistically significant at the 1% significance level. The coefficient of −0.617 (Model IV) implies that after ruling out all possible alternative explanations, the PIU caused by the GAAR announcement has a more significant adverse effect on the trading value of the treatment firm compared to the control firms in the post–GAAR period. Note that the DiD effect of Table 4 is similar in size to that reported in Table 3; the lowest value is −0.532 (Model II), which is not materially different from the

Table 4 Difference–in–differences analysis

Panel A: Mean difference–in–differences					
	Pre– GAAR I	Post–GAAR II	Difference II – I	*t*–stat	*p*–value
Treatment group	0.402	–0.164	–0.566***	–12.60	0.000
Control group	–0.463	–0.316	0.147	1.34	0.179
The difference in differences =			0.713***	–7.32	0.000
Panel B: Difference–in–differences regression					
		I	II	III	IV
		–0.659***	–0.532***	–0.548***	–0.617***
		(–5.34)	(–5.75)	(–2.84)	(–3.29)
Firm return			0.215***	0.234***	0.234***
			(15.65)	(13.32)	(13.37)
NSE return				0.091*	–0.073
				(2.00)	(–1.18)
NSE volatility				–1.438	–0.473
				(–1.36)	(–0.50)
USD volatility				–2.371	0.337
				(–0.86)	(0.10)
Real GDP growth rate				–0.706	–0.353
				(–1.55)	(–0.71)
EM return					–0.105
					(–1.04)
World return					0.176
					(1.38)
U.S. TB return					7.374**
					(2.30)
Global VIX return					0.009
					(0.75)
Firms fixed effect		Yes	Yes	Yes	Yes
Time–fixed effect (days)		Yes	Yes	Yes	Yes
Number of firms		832	832	832	832
Number of observations		47,439	47,439	47,439	47,439
Adjusted R^2 (within)		0.005	0.015	0.016	0.018
Adjusted R^2		0.143	0.152	0.153	0.154

Panel A shows the difference in differences of treatment and control groups for the average value of $NTSMC_{it}$ between the Pre–GAAR period (January 1 2012 – March 16 2012) and the Post–GAAR period (March 19 2012 – June 29 2012). $NTSMC_{it}$ is the day t net trading value by all FII scaled by the previous day's market capitalization (in basis points) of listed stocks (i) on the Indian stock market. *Treatment group* include firms having greater than the median value of the net trading sector exposure combined in January and February 2012 and the *Control group* otherwise (see Appendix C Table 11). Panel B reports the results of four different specifications of the following specification:

$$NTSMC_{it} = \beta\,(GAAR\,effect\ \times\ TRMT_GRP) + X.\delta_t + \gamma_i + \delta_t + e_{it}$$

$GAAR\,effect$ is the dummy variable, which takes the value of one for the GAAR Uncertainty Period (March 19 2012 – June 29 2012). $TRMT_GRP$ is the dummy for the firms in the *Treatment group* . $\boldsymbol{X}$ reflects the set of control variables. All the variables are defined in Appendix I. γ_i and δ_t is the vector of firm and time (day) dummies and controlling for firm and time–fixed effects, respectively. e_{it} is the error term. Standard errors are corrected for double clustering at the firm and time levels. *, ** and *** denote statistical significance at the 10%, 5% and 1% significance level respectively. The sample period ranges from January 1 – June 29 2012.

DiD effect of Table 4. The sign, statistical significance, and economically significant DiD impact support our argument that PIU generates potential deadweight costs for the FII, who react by withdrawing from the market.

Among the controls, the effect of IA is, as expected, significant, as indicated by the firm–level past returns in the Indian stock market. There is also some indication of the effects of global shocks, as noted via the statistical significance of U.S. TB return coefficients.

5.4 Propensity score matched difference–in–difference analysis

In this section, we eliminate the concern that the impact of policy uncertainty on FII net equity trading may be due to the differential firm characteristics by performing a propensity score matching (PSM) to identify a matched set of treatment and control firms.[13] To do so, we first estimate a probit model in which the dependent variable equals one if the firm belongs to the treatment group and zero otherwise. We use various firm–level characteristics, such as Tobin's Q, Total assets, ROA, Leverage, and Current liability ratio, as comparable factors (Col and Sen 2019) at the end of fiscal year 2012 (i.e., March 2013). These covariates are included to help satisfy the parallel trend assumption, as there should be no firm–specific differences in characteristics between the treatment and control groups before the policy uncertainty. Model I of Table 5 (Panel A) presents the probit model estimates with standard errors clustered at the industry level. The specification shows that some independent variables are statistically significant, suggesting significant variation in firms' characteristics between the treatment and control groups. We then use the propensity scores from Model I to perform nearest–neighbor PSM within a 0.01 caliper and end up with 144 unique pairs of matched firms.

We conduct a few diagnostic tests to verify the matching process. First, as shown in Model II in Panel A, we rerun the probit model for the propensity score–matched firms only. We found no statistically significant covariates, suggesting no observable difference in firm characteristics between the treatment and control groups. Second, we examine the difference in the propensity scores in Panel B, which shows a minimal difference (within 0.01 caliper) in the scores. Finally, Panel C reports the univariate summary of covariates between the treatment and the control group and their corresponding *t*–statistics, which do not reveal any significant mean difference in the firms' characteristics.

In Panel D, we rerun Eq. 10 (similar to Panel B of Table 4) for our matched treatment and control firms. We still find a significant impact of GAAR–related policy uncertainty on FII's net equity trading for treatment firms compared to the control firms. Even with a reduced sample size, we find a statistically significant and economically large impact ($\beta = -0.909$, $p < 0.01$), consistent with our main results.

[13] Our identification strategies classify 620 firms as treatment firms and 190 firms as control firms in our final sample.

Table 5 Propensity score matched difference–in–differences

Panel A: Pre–match propensity score regression and post–match diagnostic analysis

	Dummy = 1 if the treatment group; 0 if in the control group	
	Pre–match	Post–match
	I	II
Ln (Total assets)	0.1645**	–0.083
	(2.59)	(–0.83)
Tobin's Q	–0.057**	–0.078
	(–2.20)	(–1.25)
ROA	1.616	0.306
	(1.04)	(0.15)
Leverage	0.005	–0.030
	(0.21)	(–0.96)
Current liability ratio	–0.003	0.013
	(–0.43)	(1.17)
Industry fixed effect	Yes	Yes
Pseudo R^2	0.050	0.016
Number of observations	676	288

Panel B: Estimated propensity score distributions

	Firms	Average	Minimum	5pct	Median	75pct	Max
Control	144	0.757	0.508	0.718	0.760	0.799	0.926
Treatment	144	0.748	0.506	0.709	0.750	0.790	0.917
Difference		0.009	0.002	0.010	0.009	0.010	0.010

Panel C: Difference in firm's characteristics

	Treatment	Control	Difference	*t*–stat
	I	II	II – I	
Firm Size	9.301	9.575	–0.274	–1.61
Tobin's Q	1.644	1.917	–0.274	–1.48
ROA	0.060	0.059	0.001	0.20
Leverage	0.901	1.097	–0.196	–0.89
Current liability ratio	2.145	1.679	0.466	0.84

Panel D: PSM–Difference–in–difference analysis

	I	II	III	IV
$GAAR\,effect \times TRMT_GRP$	–0.812***	–0.695***	–0.834***	–0.909***
	(–3.68)	(–3.18)	(–3.03)	(–3.34)
Firm return		0.205***	0.239***	0.238***
		(8.45)	(10.17)	(10.21)
NSE return			0.152**	–0.075
			(2.63)	(–1.16)
NSE volatility			–0.953	0.583
			(–0.96)	(0.59)
USD volatility			–6.589**	–1.830
			(–2.35)	(–0.57)
Real GDP growth rate			–1.180**	–0.647
			(–2.63)	(–1.40)
EM return				–0.170*
				(–1.80)
World return				0.094
				(0.76)
U.S. TB return				11.75***

Table 5 (continued)

Panel A: Pre–match propensity score regression and post–match diagnostic analysis				
				(3.15)
Global VIX return				0.006
				(0.44)
Firm fixed effect	Yes	Yes	Yes	Yes
Time fixed effect (days)	Yes	Yes	Yes	Yes
Number of firms	288	288	288	288
Number of observations	14,812	14,812	14,812	14,812
Adjusted R^2(Within)	0.004	0.013	0.016	0.019
Adjusted R^2	0.163	0.171	0.174	0.175

The table reports the results of Propensity Score Matching. *Treatment group* include firms having greater than the median value of the net trading sector exposure combined in January and February 2012 and the *Control group* otherwise (see Appendix C Table 11). We use PSM with the nearest neighbourhood of 0.01 caliper using various firm–level characteristics to identify matched control groups. Panel A presents the parameter estimates from the probit model used to estimate the propensity scores for the treatment and control groups. The dependent variable is one if in the treatment group and 0 if in the control group. We include firm size, Tobin's Q, ROA, Leverage, and current liability ratio as covariates (defined in Appendix B Table 10). We include industry–fixed effects, and standard errors are corrected for clustering at the industry level. Panel B reports the distribution of estimated propensity scores post–matching. Panel C reports the univariate comparison between the treatment and control firm's characteristics and their corresponding *t*–statistics. Panel D reports PSM–difference–in–difference regression results similar to Panel B of Table 2. *, **, and *** denote statistical significance at the 10%, 5% and 1% significance levels, respectively. The sample period ranges from January 1 – June 29 2012.

6 Robustness tests

6.1 Addressing systematic shocks

Our results may capture other systematic shocks that occurred during the PIU period. If such shocks are not controlled in Model IV of Table 4, the GAAR uncertainty period factor could be confounded. We searched all the major national and international financial press to identify any significant systematic shocks that could alter the trading behavior of FII during the GAAR uncertainty period. Between March 19 and June 29 2012, we could not identify any other major unexpected shock that could drive our results. However, to rule out any such effect, we augment our regression by including the $(\boldsymbol{\alpha}_k \times \boldsymbol{\delta}_t)$ term in Eq. 10. This interaction term captures any systematic shock that could correlate with the GAAR effect (Vig 2013). Model I in Table 6 shows that this does not change the main results reported in Table 4 and remains robust to the effects of any other systematic shocks.

6.2 Alternative uncertainty period

We argue that PIU, to some extent, was removed when the Indian government made the second announcement on May 7, 2012, to postpone GAAR by an additional year. This could have provided strong signals to the FII community regarding the Indian government's intention to reduce the PIU caused by the initial GAAR announcement. In June 2012, net trading returned to positive figures, reversing the net outflows of the preceding two months. If this is the case, our GAAR Uncertainty Period (March 19 – June 29 2012) used in the previous

Table 6 Robustness tests

	Other systematic shocks	Alternative uncertaintyperiod	Balancedpanel	Alternative treatment group	
				33rd pct	Mean value
	I	II	III	IV	V
	–0.595***	–0.638***	–0.871**	–0.677***	–0.605***
	(–3.06)	(–3.45)	(–2.14)	(–3.33)	(–3.86)
Firm return	0.235***	0.261***	0.390***	0.232***	0.233***
	(13.95)	(11.71)	(9.65)	(11.77)	(13.37)
NSE return	0.072	0.000	0.160	0.075	0.073
	(1.14)	(0.01)	(1.27)	(1.14)	(1.18)
NSE volatility	–0.473	–0.704	–0.915	–0.350	–0.436
	(–0.49)	(–0.26)	(–0.54)	(–0.37)	(–0.46)
USD volatility	0.462	–2.074	4.457	0.705	0.683
	(0.14)	(–0.64)	(1.02)	(0.22)	(0.20)
Real GDP growth rate	–0.349	–0.442	0.180	–0.260	–0.429
	(–0.70)	(–0.93)	(0.25)	(–0.54)	(–0.87)
EM return	–0.108	–0.079	–0.301	–0.111	–0.096
	(–1.06)	(–0.50)	(–1.48)	(–1.09)	(–0.96)
World return	0.184	0.162	0.618**	0.219	0.176
	(1.43)	(1.31)	(2.41)	(1.67)	(1.38)
U.S. TB return	7.371**	6.923*	1.626	7.992**	6.936**
	(2.28)	(1.75)	(0.28)	(2.37)	(2.21)
Global VIX return	0.010	0.019	0.029	0.013	0.009
	(0.81)	(1.68)	(1.34)	(1.03)	(0.74)
Firms fixed effect	Yes	Yes	Yes	Yes	Yes
Time–fixed effect (days)	No	Yes	Yes	Yes	Yes
Time × Industry fixed effect	Yes	No	No	No	No
Number of firms	832	773	85	684	832
Number of observations	47,360	32,637	10,540	40,427	47,439
Adjusted R^2 (within)	0.017	0.018	0.043	0.018	0.018
Adjusted R^2	0.148	0.167	0.092	0.151	0.154

This table reports the results of five different specifications of the following equation:

$$NTSMC_{it} = \beta\,(GAAR\,effect \times\ TRMT_GRP) + \boldsymbol{X}.\delta + \boldsymbol{\gamma}_{\boldsymbol{i}} + \boldsymbol{\delta}_{t} + \boldsymbol{\alpha}_{k} \times\ \boldsymbol{\delta}_{t} + e_{it}$$

All the variables are as previously defined in Table 3, and Appendix B Table 10 notes. In Model I, we include $\boldsymbol{\alpha}_{k} \times\ \boldsymbol{\delta}_{t}$ i.e. sector × time fixed effects to control for other systematic shocks during our sample. In Model II, we use the end of May 7, 2012, as the end of the uncertainty period. In Model III, we use a balanced panel. In Model IV, we use the top (bottom) 33rd percentile value of the net trading sector exposure combined in January and February 2012. In Model V, we use the mean value of the net trading sector exposure combined in January and February 2012. *, ** and *** denote statistical significance at the 10%, 5% and 1% significance level respectively.

regressions could incorporate the reduced uncertainty period of May 8 – June 29 and thus may not fully reflect a PIU period.

To address this issue, we regenerate the uncertainty dummy from March 19 – May 7 and use this in the fully specified regression of Eq. 10. As shown in Model II of Table 6, the

DiD effect has a negative sign and is statistically significant at the 1% level. The economic impact seems to have increased (and is consistent) compared to that reported in Table 4. This is not unexpected, as the new GAAR dummy now only incorporates the period with negative outflows (see Fig. 2).

6.3 Dealing with attrition bias

Despite using an unbalanced panel dataset in the previous estimations, if the non–trading of individual firms on some dates relative to others is systematic, our estimates could suffer from sample selection or attrition bias. We, therefore, run the same fully specified regression of Eq. 10 for the same sample period of January 1 – June 29 2012, but only using a balanced panel dataset for that period. The results reported in Model II of Table 6 indicate that even after controlling for the possibility of attrition bias, our support for the PIU's impact on FII trading still holds.[14]

6.4 Alternative treatment and control groups

Our segregation of treatment and control groups based on median traded exposure can create noise for the firms very close to the cut–off point. To mitigate the impact of this possibility, we use the top 33rd percentile figure (ranked by total net trading exposure, as shown in Appendix C Table 11) as the treatment group and the bottom 33rd percentile as the control group. We also take the mean value as the cut–off value to identify the treatment and control groups. Using this reduced sample, we run a fully specified specification of Eq. 10. Our results in Model IV of Table 6, using the 33rd percentile, and in Model V, using the mean value, are virtually unchanged in terms of the sign, statistical, and economic significance of the estimations, compared to the results in Table 4.

We also define alternative treatment and control groups based on firm–level net equity trading exposure measures. We calculate the sum of each firm's FII net equity trading figures for January and February 2012 to identify the cumulative net equity trading exposure over the past two months prior to the GAAR announcement. We use the median net trading exposure figure to create the alternative firm–level treatment and control groups.

We report the mean differences in $NTSMC_{it}$ for the alternative treatment and control groups before and after the GAAR announcement in Panel A of Table 7. The qualitative results are similar but quantitatively higher in magnitude than those reported in our main findings, as shown in Table 4. The mean DiD is -2.129 basis points per day of market capitalization for a typically listed firm on the Indian stock market during the PIU period. Economically, this translates into a daily withdrawal of INR 27.08 million for each equity (around USD 0.50 million).

We then rerun Eq. 10 using alternative treatment and control groups. The results are reported in Panel B of Table 7. Consistent with our main results, the table shows that the $GAAR\,effect \times TRMT_GRP$ variable enters all regressions with an expected negative sign and is statistically significant at the 1% significance level. The coefficient of -1.648 (Model IV) implies that after ruling out all possible alternative explanations and

[14] Please note that the size of the estimates changes as the regression is based on a much smaller sample size of 10,540 observations compared to roughly 47,000 observations in Table 4. In fact, the quantitative effect seems to become even more prominent.

Table 7 Firm–level alternative treatment and control groups

Panel A: Mean difference–in–differences					
	$NTSMC_{it}$				
	Pre– GAAR I	Post–GAAR II	Difference II – I	*t*–statistics	*p*–value
Treatment group	1.362	0.116	–1.246***	–25.92	0.000
Control group	–1.527	–0.644	0.883***	13.17	0.000
The difference in differences =			–2.129***	–26.43	0.000
Panel B: Difference–in–differences regression					
		I	II	III	IV
$GAAR\,effect \times\ TRMT_GRP$		–1.266***	–1.144***	–1.584***	–1.648***
		(–10.49)	(–12.99)	(–10.03)	(–10.55)
Firm return			0.209***	0.238***	0.239***
			(15.15)	(13.59)	(13.62)
NSE return				–0.104**	–0.0719
				(–2.08)	(–1.10)
NSE volatility				–1.714	–0.527
				(–1.60)	(–0.54)
USD volatility				–4.336	–0.854
				(–1.60)	(–0.27)
Real GDP growth rate				–0.461	–0.0343
				(–1.00)	(–0.07)
EM return					–0.130
					(–1.15)
World return					0.172
					(1.26)
U.S. TB return					8.918**
					(2.68)
Global VIX return					0.00943
					(0.74)
Firms fixed effect		Yes	Yes	Yes	Yes
Time–fixed effect (days)		Yes	Yes	Yes	Yes
Number of firms		730	730	730	730
Number of observations		46,492	46,492	46,492	46,492
Adjusted R^2 (within)		0.0137	0.0237	0.0267	0.0283
Adjusted R^2		0.142	0.151	0.153	0.155

Panel A of this table shows the difference in differences of alternative treatment and control groups for the average value of $NTSMC_{it}$ between the Pre–GAAR period (January 1 2012 – March 16 2012) and the Post–GAAR period (March 19 2012 – June 29 2012). $NTSMC_{it}$ is the day *t* net trading value by all FII scaled by the previous day's market capitalization (in basis points) of listed stocks (*i*) on the Indian stock market. The alternative *Treatment group* include firms having greater than the median value of the firm–level net trading exposure combined in January and February 2012 and the *Control group* otherwise. Panel B reports the results of four different specifications of the following specification:

$$NTSMC_{it} = \beta\,(GAAR\,effect \times\ TRMT_GRP) + \boldsymbol{X}.\delta + \boldsymbol{\gamma}_{\boldsymbol{i}} + \boldsymbol{\delta}_{t} + e_{it}$$

$GAAR\,effect$ is the dummy variable, which takes the value of one for the GAAR Uncertainty Period (March 19 2012 – June 29 2012). $TRMT_GRP$ is the dummy for the firms in the alternative *Treatment group* as discussed above . $\boldsymbol{X}$ reflects the set of control variables. All the variables are defined in Appendix I. γ_i and δ_t is the vector of firm and time (day) dummies and controlling for firm and time–fixed effects, respectively. e_{it} is the error term. Standard errors are corrected for double clustering at the firm and time levels. *, ** and *** denote statistical significance at the 10%, 5% and 1% significance level respectively. The sample period ranges from January 1 – June 29 2012.

using alternative firm–level treatment and control groups, the PIU caused by the GAAR announcement has a more significant adverse effect on the equity trading value of the treatment firm compared to the control firms in the post–GAAR period.

6.5 Addressing the possibility of a cyclical effect

Finally, although there is no economic justification, it is possible that our results are simply capturing cyclical effects that may be prevalent in the Indian stock market from March 19 to June 29 2012, every year. If this is the case, we may be exposed to the possibility of a false experiment by capturing the regular cyclical effect. To rule out this possibility, we run placebo effect regressions capturing the same period of March 16 – June 29 for the same treatment and control groups but for the years 2010 (two years before the PIU) and 2014 (two years after the PIU). For this, we create two dummies, taking the value of one if the period covered is March 19 – June 29 for both 2010 and 2014 (i.e., placebo years). We interact these dummies with our treatment dummy, denoting them as a *Cyclical effect* in our regressions.[15] If our regression captures any cyclical impact, the estimates would be similar in size, sign, and statistical significance to those reported for 2012. Table 8 shows that the estimate for 2010, although statistically insignificant, carries the opposite sign of the GAAR effect. The estimate for the year 2014 is also positive and statistically insignificant, providing confidence that cyclical effects do not drive our results.

6.6 Other alternative explanations

Given that the systematic change in trading behavior during the uncertainty period could arguably be associated with PIU, we need to add caveats and rationales to rule out other possible explanations. First, our results, consistent with Ülkü (2015) and Sarno et al. (2016), generally find that recent equity returns and global push factors (particularly U.S. TB return) also play a significant role in driving portfolio flows. However, most existing studies examine the effect of different lag structures of local and global factors. When we include lagged values of up to five periods of the controls, the results related to the DiD effect remain statistically and economically similar. We do not report the results for brevity; however, they are available from the authors upon request.

Another possible argument is whether the change we observe in FII trading is due to the first announcement itself or attributed to the uncertainty it created. There are two possible reasons why this may not be the case. First, the announcement per se does not change any existing tax rate in absolute economic terms; instead, it offers tax authorities greater rights to securitize tax avoidance transactions of FII. Second, even if we assume that, in an equilibrium framework, investors undertake a reallocation with the new information, it should occur within a short period, rather than, as we typically find, until the uncertainty is resolved. Another potential driver could be the effect of the overall market price index, where GAAR does not affect the fundamental value of the stocks but alters payoffs to only one investor group, i.e., FII. From the point of view of an accurately calibrated theoretical equilibrium model, the new equilibrium price could be an essential factor in trading FII, mainly because FII could buy the index when it is below its fundamental value. However,

[15] Note these are exactly the same regressions as specified in Eq. 10 (fully specified), including the same sample period but for the years 2011 and 2013 separately.

Table 8 Addressing cyclical effect: false experiments

	Year 2010	Year 2014
	I	II
$Cyclical\ effect$	0.420	0.166
	(1.62)	(0.82)
Firm return	0.337***	0.150***
	(9.95)	(9.30)
NSE return	0.389***	0.0252
	(2.78)	(0.34)
NSE volatility	–0.259	–4.212***
	(–0.33)	(–4.95)
USD volatility	1.392	–3.215**
	(0.73)	(–2.06)
Real GDP growth rate	2.006***	1.703**
	(3.00)	(2.18)
EM return	0.202	0.168
	(1.25)	(1.58)
World return	0.143	–0.261
	(0.80)	(–1.44)
U.S. TB return	1.330	3.144
	(0.45)	(0.52)
Global VIX return	0.013	–0.021*
	(0.75)	(–1.84)
Firms fixed effect	Yes	Yes
Time–fixed effect (days)	Yes	Yes
Number of firms	848	830
Number of observations	36,423	44,213
Adjusted R^2 (within)	0.012	0.012
Adjusted R^2	0.149	0.161

This table reports the regression results of the following specification run for the years 2010 and 2014:

$$NTSMC_{it} = \beta\ (Cyclical\ effect) + \boldsymbol{X}.\delta + \boldsymbol{\gamma}_{\boldsymbol{i}} + \boldsymbol{\delta}_{t} + e_{it}$$

All the variables are as previously discussed in notes to Table 3, and Appendix I. *Cyclical effect* is the interaction of a dummy created for the period March 16 – June 29 for the *Year 2010* and *Year 2014* models with the treatment group ($TRMT_GRP$). The period is the same as the *GAAR* dummy for 2012 in the previous regressions. $TRMT_GRP$is the dummy for the treatment group of firms having greater than the median value of the net trading sector exposure combined for January and February for the years 2011 and 2013, respectively. $\boldsymbol{X}$reflects the set of control variables defined in Appendix B Table 10. $\boldsymbol{\gamma}_{\boldsymbol{i}}$and $\boldsymbol{\delta}_{t}$ is the vector of firm and time dummies controlling for firm and time–fixed effect. $e_{it}$is the error term. Standard errors are corrected for double clustering at the firm and time (day) levels. *, ** and *** denote statistical significance at the 10%, 5% and 1% significance level respectively. The sample period ranges from January 1 – June 29 of 2010 in Model I and 2014 in Model II.

we have controlled for this using both the stock–level return index and the market–level return index. Thus, any price pressure factors are considered in the empirical setup.

Finally, other firm characteristics such as corporate governance, gearing, return on equity, and book–to–market value could also drive the results. However, the data available on these firm–level characteristics for our period are those of March 31, 2012, for each firm. Since we only have one observation for each firm for the entire sample period, we cannot use these data in our fixed–effect model. Hence, any firm–specific effect is accounted for in

the estimations. Moreover, the firm's previous day's return significantly captures any time–varying firm–specific effects.

6.7 Post–GAAR eradication period

The theoretical framework discussed in Sect. 3 suggests that FII equity trading should be lower during the post–GAR period. However, it is important to consider the FII's trading reaction following the announcement on June 29, 2012, which removed the uncertainties surrounding GAAR. First, we conduct a simple paired *t*–test for the mean differences in $NTSMC_{it}$ before and after the announcement on June 29, 2012, using six different window periods. The post–GAR period denotes trading days before the third announcement on June 29, 2012, and the post–GAR eradication period denotes trading days after the third announcement. We also conduct a short–term and long–term window period analysis to investigate whether FII reversals continue in the long term. For the short–term window period, we investigate the trading period between March 19, 2012, and September 28, 2012 (*GAAR Eradiation SP*). For the long–term window period, we investigate the trading period between March 19, 2012, and March 31, 2013 (*GAAR Eradication LP*).

We report the result in Table 9. In Panel A, we find a significant rise in $NTSMC_{it}$ in all post–GAAR eradication window periods (except for the 1–week window period) compared to the post–GAAR period. It is also important to note that, though there is a rise in $NTSMC_{it}$ in short window periods, the level of $NTSMC_{it}$ is very low compared to $NTSMC_{it}$ in the pre–GAAR period. For instance, the average $NTSMC_{it}$ In the pre–GAAR period, there are 0.343 basis points (Table 1), whereas it is around 0.072 basis points in the short–term post–GAAR eradication period. However, average $NTSMC_{it}$ in *GAAR Eradication LP* is 0.316 basis points, similar to the pre–GAAR period. These results suggest that while uncertainties result in an immediate outflow of FII equity trading, removing uncertainties does not necessarily result in an immediate inflow. These findings are consistent with the lingering effect of uncertainty discussed in H_2.

In Panel B, we investigate the FII equity trading using the following regression equations:

$$NTSMC_{it} = \beta_1 (GAAR\ Eradication) + \boldsymbol{X}.\delta + \boldsymbol{\gamma_i} + e_{it} \tag{11}$$

$$NTSMC_{it} = \beta_1 (GAAR\ Eradication \times TRMT_GRP) + \boldsymbol{X}.\delta + \boldsymbol{\gamma_i} + \boldsymbol{\delta_t} + e_{it} \tag{12}$$

where, all the variables are as previously defined. $GAAR\ Eradication$ is a dummy variable which takes the value of zero for the post–GAAR period (March 19 – June 29, 2012) and one for the post–GAAR eradication period (July 2 – March 28, 2013) for different window periods as specified previously. Models I – V results reveal findings consistent with those of Panel A. However, once we introduce the push and pull factors in Model VI, we do not find any significant increase in $NTSMC_{it}$ in the short–term post–GAAR eradication period compared to the post–GAAR period. We also do not find any significant change in our DiD setting in Model VII, which reveals that the shift in $NTSMC_{it}$ in the post–GAAR eradication period for the treatment group firms is not significantly different from the change in $NTSMC_{it}$ for the control group firms in the short–term. These results suggest that removing PIU does not necessarily result in an immediate inflow of FII equity trading. These subdued flows are consistent with the lingering effect of uncertainty literature

Table 9 Post–GAAR uncertainty

Panel A: Daily average net equity trading									
Window periods			Post–GAAR	Post–GAAR Eradication		Diff	*t*–stat	*p*–value	Observations
1–Day			–0.759	0.121		0.880	2.851***	0.004	766
1–Week			0.098	0.180		0.081	0.586	0.558	3773
1–Month			–0.215	0.092		0.308	4.893***	0.000	16,191
2–Months			–0.267	0.053		0.320	7.195***	0.000	32,381
GAAR Eradication SP			–0.200	0.072		0.272	7.363***	0.000	50,781
GAAR Eradication LP			–0.200	0.316		0.516	16.292	0.000	100,185
Panel B: Regression analysis in GAAR eradication period									
	One–day window	One–week window	One–month window	Two–month Window	GAAR Eradication Short Period	GAAR Eradication Short Period	GAAR Eradication Short Period	GAAR Eradication Long Period	GAAR Eradication Long Period
	I	II	III	IV	V	VI	VII	VII	IX
$GAAR\ Eradication$	1.105***	0.200	0.169*	0.202**	0.194**	–0.344		0.277**	
	(3.40)	(1.27)	(1.77)	(2.46)	(2.41)	(–1.01)		(2.07)	
$GAAR\ Eradication \times TRMT_GRP$							–0.110		0.220**
							(–0.58)		(2.28)
Firm return						0.235***	0.210***	0.205***	0.160***
						(13.41)	(12.41)	(17.25)	(12.78)
NSE return						–0.073	0.020	0.096***	0.102**
						(–1.19)	(0.34)	(4.26)	(2.57)
NSE volatility						–0.468	–0.157	–0.419	–0.879*
						(–0.48)	(–0.22)	(–1.50)	(–1.89)
USD volatility						0.393	2.528	–0.195	–0.457
						(0.13)	(1.02)	(–0.38)	(–0.70)
Real GDP growth rate						–0.356	–0.081	0.171	0.149
						(–0.65)	(–0.18)	(0.37)	(1.42)
EM return						–0.104	–0.048	0.022	0.038
						(–1.03)	(–0.54)	(0.94)	(0.61)

Table 9 (continued)

World return						0.176	0.128	0.113***	0.116*
						(1.37)	(1.47)	(3.21)	(1.76)
1–Year U.S. TB return						7.350**	3.727	3.783**	3.449
						(2.20)	(0.68)	(2.45)	(0.99)
Global VIX return						0.009	–0.002	–0.001	–0.001
						(0.74)	(–0.33)	(–0.34)	(–0.04)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effect (days)	No	No	No	No	No	Yes	Yes	Yes	Yes
Number of firms	298	510	702	799	874	874	869	1.093	1,093
Observations	596	3,647	16,095	32,273	50,684	50,684	50,684	97,024	97,024
Adjusted R^2 (within)	0.038	0.001	0.001	0.001	0.001	0.012	0.012	0.012	0.013

Panel A is similar to Table 1. For the one–day window, the *Post–GAAR Period* is June 28, and the *Post–GAAR Eradication* period is July 2 2012. The pre–window of *Post–GAAR Period* ranges from March 19 – June 29 2012, and the post–window of *Post–GAAR Eradication Period SP* ranges from July 2 – September 28 2012 for short–term period and *Post–GAAR Eradication Period LP* ranges from July 2 2012 to March 29 2013 for long–term period. Panel B reports the results of different specifications of the following equations:

$$NTSMC_{it} = \beta_1 \times GAAR\ Eradiction + X.\delta + \gamma_i + e_{it}$$

$$NTSMC_{it} = \beta_1 (\ GAAR\ Eradiction \times TRMT_GRP) + X.\delta + \gamma_i + \delta_t + e_{it}$$

$GAAR\ Eradication$ is the dummy variable which takes the value of one for the post–GAAR period and zero for the post–GAAR eradication period. $TRMT_GRP$ is the dummy for the firms in the *Treatment group*. $\boldsymbol{X}$ reflects the set of control variables. All the variables are defined in Appendix I. γ_i and δ_t is the vector of firm and time (day) dummies and controlling for firm and time–fixed effect, respectively. e_{it} is the error term. Standard errors are corrected for double clustering at the firm and time levels (where specified). *, ** and *** denote statistical significance at the 10%, 5% and 1% significance level respectively

and H_2, as noted in Sect. 3 (Julio and Yook 2012; Gulen and Ion 2016; Jens 2017; Honig 2020). In Models VII and IX, we examine the trading behavior of FII over a longer window period. The impact of GAAR eradication is statistically significant and positive in both settings. Our findings reveal that the lingering effect of uncertainty subsides over a longer window period, and capital flows return to equilibrium before the PIU.

7 Concluding remarks

The literature notes that FII in emerging markets are momentum investors driven by recent returns (Brennan and Cao 1997; Griffin et al. 2004). The current literature attributes such feedback trading behavior to the theoretical notion of IA between local and foreign investors (Froot et al. 2001; Kim and Wei 2002; Griffin et al. 2004; Richards 2005). However, our study suggests that FII's trading behavior in emerging markets is not only driven by IA but also by the degree of PIU. Our study uses the unexpected announcement of imposing GAAR on the trading of FII as an example of PIU, which aims to control tax–avoiding transactions. The uncertainty in the GAAR provisions was that it was unclear whether it would be imposed retrospectively and what level of authority would be delegated to the tax officials to question transactions made by FII. However, the Indian government made another announcement on May 7 2012, postponing the effective date of GAAR by one year, and finally announced it on June 29 2012, to remove the uncertainties. Using the period from March 19 to June 29, 2012, as the PIU period and transaction–level data from the Indian stock market, we undertake a quasi–natural experiment to investigate the reaction of FII to the GAAR announcements.

Our results, robust to various alternative tests, provide three crucial findings. First, we find that FII negatively reacted to the PIU caused by the announcement of the GAAR provisions. The three months before the GAAR announcement have witnessed positive net investments from FII. However, net trading became negative in the post–GAAR announcement period. The average daily withdrawal during the uncertainty period is at least 0.713 basis points of the market capitalization for the traded equities. Second, the reaction is immediate as the boycott is observed from the next trading day following the announcement. Third, the withdrawal of funds by FII continues until the PIU is removed. Finally, the removal of PIU does not result in an immediate inflow of foreign equity trading. Our study thus concludes that PIU is an important determinant of the trading behavior of FII.

The results of our study suggest that FII demonstrated heightened sensitivity to tax policies, and any abrupt changes that raise their tax burden, especially retrospective, could induce substantial capital outflows. Policymakers in emerging markets competing for global mobile capital must ensure clarity and stability in investment–related policies, such as avoiding retrospective tax rules. While FII may exit swiftly in response to unfavorable tax changes, they exhibit reluctance to reinvest at the same pace even after PIU is reversed, which is concerning for policymakers. FII seems to delay reinvesting until they gain confidence about the reforms taken by policymakers to attract international capital. Consequently, policymakers aiming to attract and sustain FII participation in the equity market should exercise caution when designing, communicating, and executing regulations directly influencing FII projected returns, ensuring stability and confidence in market conditions.

Appendix A: GAAR background

Khanna (2014)[16] notes that historically nearly 40% of foreign portfolio investments[17] in India flow through Mauritius, the latter being a prominent place for tax planning of global fund houses investing in India. The India–Mauritius double taxation avoidance agreement (agreed in 1983) provides for the taxability of capital gains from the sale of securities in India. However, the local laws of Mauritius, which offer a minimal tax rate, make such transactions tax–free. Such an amicable tax regime has attracted global investors to Mauritius but for potentially the sake of treaty shopping. Further, Munshi (2012b) notes that most foreign investments in the Indian market are made indirectly via funds, promissory notes (P–note), or derivatives mimicking the underlying security. This allows investors to avoid Indian taxes on direct investments, which could be as high as 40%. However, India has been increasingly alarmed about routing third–country investment through Mauritius and the round–tripping of investments leading to the loss of substantial tax revenues.[18]Such loss of payments is also doubted to be routed from other jurisdictions, particularly countries considered tax havens, such as the Cayman Islands, Singapore, Hong Kong, etcThe issue of tax avoidance also came into the limelight from a controversial deal of Vodafone, a UK–based multinational telecom company, buying an indirect but controlling stake in Hutchison Essar Limited (HEL).[19] HEL was a joint venture in India with Hutchison Telecommunications International Limited (HTIL), which held and operated telecom licenses in India. In 2007, the Dutch subsidiary of Vodafone acquired a stake in HEL from a subsidiary of HTIL registered in the Cayman Islands. This deal accorded Vodafone a controlling stake of 67% of HEL, costing $11.2 billion. Since the underlying asset (i.e., HEL) was located in India, the Indian government claimed that the deal was taxable amounting to roughly $2.5 bnVodafone objected to the tax claim and went to India's Supreme Court, arguing it was a deal between two foreign entities in a foreign jurisdiction. Thus, the Indian government had no right to impose capital gains tax. On January 20 2012, the Supreme Court of India ruled in favour of Vodafone. In the aftermath, the Indian legislature began considering tax reforms to deal with the indirect transfer of shares of an Indian company, where at least 50% of the assets of the transfer (directly or indirectly) are related to investments in India. Although not expected by foreign investors, the GAAR was also the outcome of such an effort by the Indian government.[20]

[16] URL: http://www.grantthornton.in/assurance–tax–regulatory–framework/walking–the–extra–mile/ (accessed 12 October 2015).

[17] Munshi (2012a), Financial Times, points out foreign investments constitute nearly 17% of the capitalization of Indian markets (source: Asian Securities Industry and Financial Markets Association).

[18] Munshi (2012a), Financial Times, quote: "…One analyst, who did not wish to be named, gave the example of a politician routing money from a Swiss bank through a P–note via Mauritius in order to launder it, and turn it into white money eligible to be invested in Indian equities".

[19] Lamont (2011). "Vodafone warns India tax bill to hit $5bn". FT.com. Retrieved 15 June 2013.

[20] Crabtree (2012), Financial Times, notes: "But Mr. Mukherjee (Indian Finance Minister) disappointed investors by ploughing ahead with plans to allow the retrospective taxation of certain international transactions, a move in part designed to recover more than US$2 billion of capital gains tax from UK–based telecoms group Vodafone."

Appendix B

Table 10 Definition of variables

Variable	Definition
Dependent variable	
$NTSMC_{it}$	The day *t* net trading value by all FII scaled by the previous day's market capitalization (in basis points) of listed stocks (*i*) on the Indian stock market.
Main independent variable	
$GAAR\,effect$	Dummy variable that takes the value of one for the GAAR uncertainty period between March 19 2012 – June 29 2012, and zero for the period between January 1, 2012, to March 16 2012.
$TRMT_GRP$	Dummy variable that takes the value of one for the *Treatment group* and zeroes for the *Control group. Treatment group* include firms having greater than the median value of the net trading sector exposure combined in January and February 2012 and the *Control group* otherwise (see Appendix C).
Push and Pull factors	
Firm return	The previous day's return of the firm traded by FII.
NSE return	The previous day's return on the National Stock Exchange (NSE500 index)
NSE volatility	The daily standard deviation estimated using the past 90 days' return on the National Stock Exchange
USD volatility	The daily standard deviation of the USD/INR exchange rate
Real GDP growth rate	The last quarter's real GDP growth rate
EM return	The previous day's return on the MSCI emerging market index.
World return	The previous day's return on the MSCI world market index.
U.S. TB return	The previous day's return on one year U.S. Treasury bill rates
Global VIX return	The previous day's return on the Chicago Board Option and Exchange Volatility
PSM Covariates	
Ln (Total assets)	Log of total assets
Tobin's Q	The sum of the book value of debt and market value of equity divided by total assets
ROA	Return on assets calculated as net profit divided by total assets
Leverage	The ratio of total debt to total equity
Current liability ratio	The ratio of current liabilities to total liabilities

Table 11 Treatment and comparison group

Sector–based on the two–digit National Industrial Classification of India	Net equity trading value in INR million					Treatment
	Jan	Feb	Jan/Feb	March	April	
Financial service activities, except insurance and pension funding	18,439	88,561	107,001	11,921	8,509	1
Computer programming, consultancy and related activities	17,640	27,958	45,598	16,450	–6,591	1
Manufacture of motor vehicles, trailers and semi–trailers	16,647	8,610	25,257	5,907	7,944	1
Manufacture of basic metals	12,044	10,383	22,427	–3,372	–4,111	1
Manufacture of coke and refined petroleum products	4,864	16,222	21,087	–887	–8,948	1
Civil engineering	8,717	5,778	14,494	–946	–4,942	1
Manufacture of machinery and equipment	5,025	9,228	14,253	–389	–2,440	1
Extraction of crude petroleum and natural gas	4,669	8,831	13,500	–2,980	1,528	1
Manufacture of other non–metallic mineral products	5,220	6,408	11,628	3,476	897	1
Manufacture of pharmaceuticals, medicinal chemicals and botanical products	4,516	6,624	11,140	11,012	5,086	1
Manufacture of tobacco products	4,602	4,131	8,733	10,872	2,480	1
Electricity, gas, steam, and air conditioning supply	2,009	6,653	8,662	1,088	1,457	1
Manufacture of food products	1,748	4,662	6,411	2,118	–303	1
Manufacture of chemicals and chemical products	–1,735	5,753	4,018	5,282	–198	1
Manufacture of other transport equipment	–1,351	4,729	3,377	1,594	1,352	1
Construction of buildings	453	1,103	1,555	–1,157	–349	1
Repair and installation of machinery and equipment	–3,667	4,571	903	4,520	806	1
Manufacture of electrical equipment	–832	1,606	774	375	–306	1
Manufacture of fabricated metal products, except machinery and equipment	–146	817	670	–245	756	1
Manufacture of textiles	131	419	550	319	57	1
Other manufacturing	–593	1,071	478	2,163	–274	1
Mining support service activities	183	144	327	–373	–3	1
Accommodation	86	196	282	–52	24	1
Water transport	231	13	245	–15	–74	1
Mining of metal ores	143	95	238	–376	5	1
Real estate activities	–780	999	219	–751	–519	1
Mining of coal and lignite	19	170	189	–29	–63	1
Sports activities and amusement and recreation activities	184	–20	164	–36	–102	0
Human health activities	98	–26	72	6,590	–56	0
Activities of head offices; management consultancy activities	8	47	55	161	–9	0
Manufacture of rubber and plastics products	–626	676	51	–248	–485	0
Forestry and logging	–	13	13	–	48	0
Manufacture of paper and paper products	–30	40	10	48	160	0
Architecture and engineering activities; technical testing and analysis	6	0	7	6	8	0
Other financial activities	0	1	1	3	2	0

Table 11 (continued)

Sector–based on the two–digit National Industrial Classification of India	Net equity trading value in INR million					Treatment
Public administration and defence; compulsory social security	–2	–1	–3	0	56	0
Activities of membership organizations	–2	–21	–22	–7	–	0
Publishing activities	4	–28	–24	276	98	0
Air transport	–23	–11	–34	314	11	0
Travel agencies, tour operators and other reservation service activities	–63	26	–37	29	17	0
Manufacture of wearing apparel	–40	–8	–48	342	144	0
Rental and leasing activities	–23	–49	–72	–62	–	0
Advertising and market research	–9	–138	–147	–4	–1	0
Retail trade, except for motor vehicles and motorcycles	–60	–89	–148	1,362	715	0
Motion picture, video and television programme production, sound recording and music publishing	–168	13	–155	23	–105	0
Employment activities	–41	–132	–172	99	–25	0
Crop and animal production, hunting and related service activities	–165	–113	–278	26	–114	0
Information service activities	–424	144	–279	48	–92	0
Education	–211	–145	–355	–233	–70	0
Land transport and transport via pipelines	–229	–192	–421	–1,105	–980	0
Wholesale trade, except for motor vehicles and motorcycles	–755	–435	–1,191	–240	–170	0
Manufacture of computer, electronic and optical products	–843	–1,301	–2,144	–59	–411	0
Warehousing and support activities for transportation	–2,882	282	–2,600	–673	–1,409	0
Manufacture of beverages	–1,877	–1,749	–3,626	293	2,831	0
Telecommunications	–256	–5,188	–5,444	2,174	–2,140	0

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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